APL confined to its calyx sub-arbors

> **QUESTION**
>
> If signals inside APL cannot cross between its calyx sub-arbors (the compartments of Report 10), does the model reshape the KC representation, especially the rank order of odor-pair similarity with inhibition on versus off, more like TNT vs TNT3c than a uniform APL, an APL weighted by per-KC synapse counts, or the local APL without walls of Report 11?

INPUTS

- The pinned KC model of Report 07
- APL skeletons and calyx sub-arbors (Report 10)
- Calyx KC ↔ APL and APL → PN synapses with their positions (Reports 01, 02)
- The local model of Report 11, as reference

OUTPUTS

- `odor_weighted/APL_compartment_model_summary.json`, `_fits.csv`, `_per_MB_{ORN,GH146}.csv`, `_odor_pairs.csv`, `_nulls.csv`, `_compartments.csv`
- `odor_weighted/APL_compartment_model.png`, `APL_compartment_model_notes.md`
- the same in `with_outside_ROI_synapses/odor_weighted/`

SCRIPTS

- `scripts/apl_compartment_model.py` (the model)
- `scripts/apl_compartment_model_figure.py`
- `scripts/apl_compartment_model_notes.py`
- `scripts/reports/r12_prep.py` (compartment spread, matrix properties, positions)
- `scripts/reports/r12_compartments.py` (this report)

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Report 10 found that APL's calyx arbor is not one continuous tree in the calyx: it comes in several sub-arbors that join only farther down, through the peduncle or the lobes, so two synapses in different sub-arbors are 200–360 µm apart along APL even when they are close in space. Report 11 let APL's signal fade smoothly with distance along the skeleton. This report tries the other extreme suggested by that anatomy: each sub-arbor is a closed compartment, and a KC's activity can only inhibit KCs whose APL synapses sit in the same sub-arbor. Hemibrain R, male CNS L and R (Yang, 2026-10-21), both versions of the tables.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
k, j	Kenyon cells (KCs); k receives inhibition, j excites APL	N = 1,793 (hemibrain R), 1,876 (male CNS L), 1,900 (male CNS R)

Symbol	What it stands for	Values, units, range
o	An odor	16 (ORN input) or 13 (GH146 input)
$x_{ko},\ y_{ko}$	Excitatory input and response of KC k to odor o (Report 07, pinned)	threshold units; ≥ 0
θ	KC threshold	1
a	APL gain onto KCs, one per mushroom body and version, set so that the mean intact fraction responding equals TNT3c's	> 0
M_{kj}	Coupling: how strongly activity of KC j inhibits KC k through APL	mean over k, j set to that of C
c	A compartment: one connected calyx sub-arbor of APL (named S1, S2, ... in Report 10; small pieces are compartments too)	
I_{cj}	KC → APL synapses from KC j in compartment c (APL's inputs)	count
O_{ck}	APL → KC synapses onto KC k in compartment c (APL's outputs)	count
L_c	Cable length of compartment c: summed skeleton length inside it	µm
$d(s,t)$	Geodesic distance between sites s and t along APL's skeleton	µm
λ	Length constant of passive decay inside a compartment	10–200 µm, or ∞ (no decay)
κ_v	Decay factor of the tree edge from node v to its parent: $e^{-\ell_v/\lambda}$ inside a compartment, 0 on an edge that leaves one	0 to 1
Φ_{vj}	Signal from KC j at skeleton node v	≥ 0
$B_{kb},\ Q_{bo},\ \beta$	Bouton route of Report 11: share of KC k's claw input from bouton b; APL signal at bouton b; its divisive gain	
f	Boutons' share of APL's calyx output synapses	0.097 hemibrain, 0.112 male CNS
S_{ab}	Cosine similarity of the KC population responses to odors a and b	0 to 1
ρ_{rank}	Spearman correlation over odor pairs between blocked and intact S_{ab}, mean over the three mushroom bodies	−1 to 1; lower = more reordering
r	Pearson correlation over odor pairs between the model's ΔS_{ab} (blocked − intact, averaged over mushroom bodies) and the data's (TNT − TNT3c)	−1 to 1
π	A random permutation of KCs within their class (αβ, α′β′, γ), for the null	

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1 The compartments: APL's calyx sub-arbors

The compartments are exactly the sub-arbors of Report 10. Skeleton nodes within 5 µm of a calyx APL synapse (from the table with synapses outside the official ROIs added, which covers the whole calyx) are calyx nodes. Keeping only those nodes and the skeleton edges between two of them splits APL's calyx arbor into connected pieces, because the pieces are joined only through nodes outside the calyx. Each piece is one compartment. Pieces holding at least 2% of APL's synapses on microglomeruli keep Report 10's names S1, S2, ...; the rest are small pieces, and each small piece is a compartment of its own. Every calyx KC ↔ APL synapse is placed at its nearest skeleton node and so belongs to exactly one compartment (none fell outside a piece).

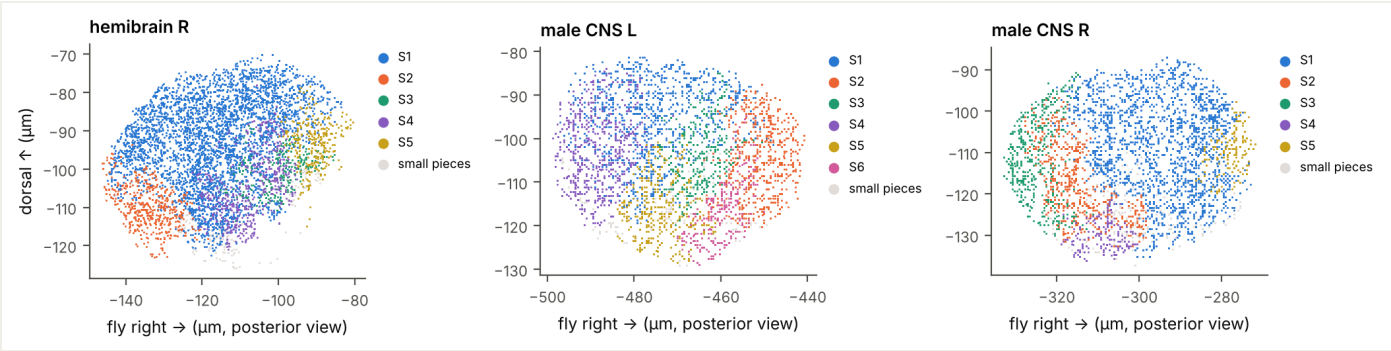


Figure 1. APL's calyx compartments. Calyx KC ↔ APL synapses (a random 8,000 per mushroom body, official ROIs), each drawn at the APL skeleton node it was placed on, viewed from posterior (x: µm toward the fly's right; y: µm toward dorsal; hemibrain and male CNS have different origins, so only distances within a panel compare). Color: the compartment (sub-arbor) of that node, named and colored as in Report 10; light gray: small pieces. Neighboring colors overlap in this 2-D view because the sub-arbors interleave in depth.

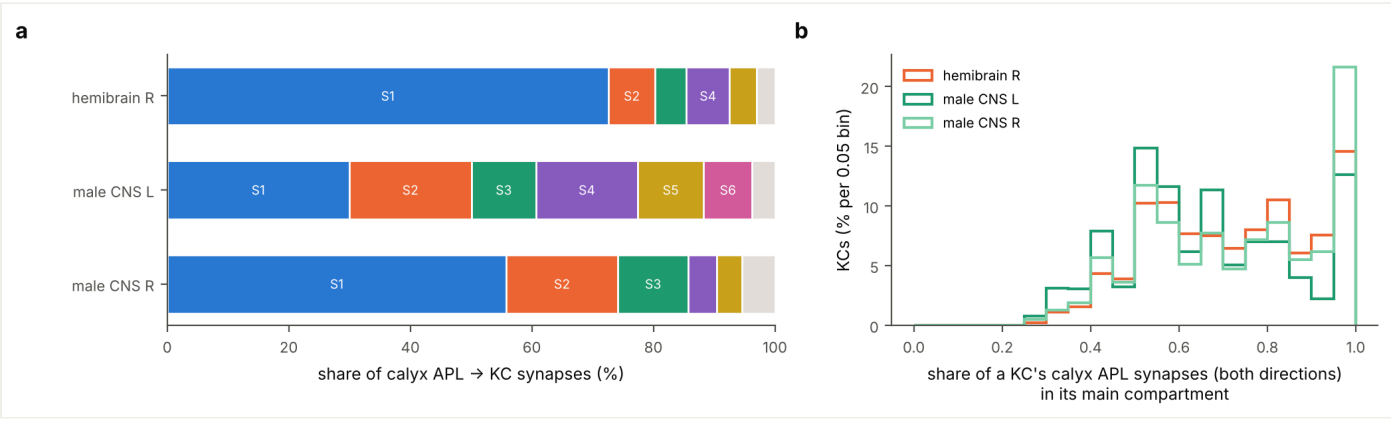


Figure 2. How APL's output and each KC's synapses divide among compartments. a, how APL's calyx output onto KCs (APL → KC synapses) divides among the compartments of each mushroom body (official ROIs); colors and names as in Figure 1, light gray = all small pieces together. **b**, for each KC, the share of its calyx synapses with APL (KC → APL and APL → KC together) that lie in the one compartment holding most of them; histogram over KCs with at least one such synapse, lines = mushroom bodies, y in % of those KCs per 0.05 bin. A value of 1 means all of a KC's APL synapses are in one compartment.

Table 2. Compartments, official ROIs. Effective number = $1/\sum_c s_c^2$ with s_c the compartment's share of APL → KC synapses (equal to the count if all were equal). The last two columns use each KC's synapses with APL in both directions.

Mushroom body	compartments with KC synapses	named (≥ 2%)	effective number	largest share of APL → KC	KC main-compartment share, median	KCs entirely in one compartment
hemibrain R	16	5	1.8	73%	0.72	12%
male CNS L	15	6	5.3	30%	0.64	11%
male CNS R	11	5	2.7	56%	0.73	20%

Hemibrain R has one dominant compartment (effective number 1.8), so walls change little there; male CNS L is split most evenly. Most KCs have their APL synapses in more than one compartment, because a KC's claws are spread over the calyx, but usually with a clear main one (median share 0.72 in hemibrain R, 0.64 in male CNS L, 0.73 in male CNS R).

2 The model, step by step

2.1 Step 1: KC input and pinning (unchanged)

As in Reports 07 and 11: each KC's input is its claws' PN synapse shares times the glomerular ORN (or GH146) responses; each odor's input gain is set so that with APL blocked the fraction of KCs responding equals that odor's TNT fraction; the blocked responses $y_{ko}^{\text{blocked}} = [x_{ko} - 1]_+$ are the same for every version.

2.2 Step 2: feedback through a coupling matrix

$$y_{ko} \;=\; \Big[\, x_{ko} - \theta - a \sum_{j} M_{kj}\, y_{jo} \Big]_+ \tag{1}$$

The steady state of KC responses with APL intact, solved per odor. Versions differ only in M.

Isopotential compartments ($\lambda = \infty$). Inside one compartment the APL potential is the same everywhere; nothing passes between compartments. KC j raises the potential of compartment c in proportion to its KC → APL synapses there, I_{cj}, and KC k is inhibited by that potential at its APL → KC synapses there, O_{ck} :

$$M_{kj} \;=\; \text{const} \times \sum_{c} O_{ck}\, I_{cj} \tag{2}$$

A sum over compartments of per-compartment "C" matrices. With a single compartment this is exactly C.

Isopotential, divided by cable length ("÷ length"). A passive compartment of membrane area A turns an input current into a voltage change in proportion to its input resistance, which scales as $1/A$. Taking area proportional to cable length L_c (same branch thickness), the same input moves a small compartment's potential more than a large one's:

$$M_{kj} \;=\; \text{const} \times \sum_{c} \frac{O_{ck}\, I_{cj}}{L_c} \tag{3}$$

L_c in µm; a piece with no edge counts as 1 µm.

Decay inside compartments (λ = 200, 100, 50, 25, 10 µm). The local model of Report 11, with every skeleton edge that leaves a compartment cut:

$$M_{kj} \;=\; \text{const} \times \sum_{s \,\in\, \text{out}(k)} \; \sum_{t \,\in\, \text{in}(j)} \mathbf{1}\big[\, c(s) = c(t) \,\big]\; e^{-d(s,t)/\lambda} \tag{4}$$

$\text{out}(k)$: APL → KC sites on KC k; $\text{in}(j)$: KC → APL sites of KC j; $c(s)$: compartment of site s.

In each version the constant sets the mean of M to that of C; the gain calibration (Step 4) would absorb it anyway.

2.3 Step 3: computing M on the tree with walls

The two-pass computation of Report 11 (pass 1 from the leaves up, pass 2 from the root down, each edge v weighted by κ_v) gives $\Phi_{vj} = \sum_t e^{-d(v,t)/\lambda}$ exactly on a tree. Walls are added by setting $\kappa_v = 0$ on every edge whose two ends are not in the same compartment; a signal can then never cross that edge. Because the path between two nodes of one connected piece stays inside the piece, an edge of the reduced tree that joins two nodes of the same compartment never leaves it, so this test on the reduced tree is exact. For $\lambda = \infty$ (κ_v = 1 inside), the result was compared with $\sum_c O_c^\top I_c$ computed directly: identical in all three mushroom bodies (difference exactly 0).

2.4 Step 4: gains, solver, bouton route

For every mushroom body, input table and version, the gain a is found by bisection so that the mean fraction of KCs responding equals TNT3c's (6.83% for the 16 odors), with the exact active-set solver of Report 11. The bouton route of Report 11 is kept, with the APL signal at each bouton taken from the same walled computation (Q_{bo} from Φ at its APL → PN synapses): **boutons only** (all inhibition on boutons) and **KCs + boutons** at the boutons' real share f of APL's calyx output, for λ = ∞ and 50 µm.

2.5 Step 5: readouts and null

- **Rank order** ρ_{rank}, **pattern** r and **size** (mean ΔS), exactly as in Report 11. Data: ρ_{rank} = 0.869 (ORN odors) and 0.873 (GH146 odors), about 0.92 after correcting for recording noise; r can reach about 0.84 at most; mean ΔS 0.178 and 0.201.
- **Null**: for the isopotential version (∞ or ∞ ÷ length) with the better r, KCs swap their APL synapse sites with other KCs of the same class (KC k gets KC $\pi(k)$'s sites, $M \to M_{\pi(k)\pi(j)}$), 100 times, recalibrating each time. This asks whether it matters which KC sits in which compartment. One-sided p = (1 + shuffles at least as extreme) / 101.

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3 What walls do to the coupling

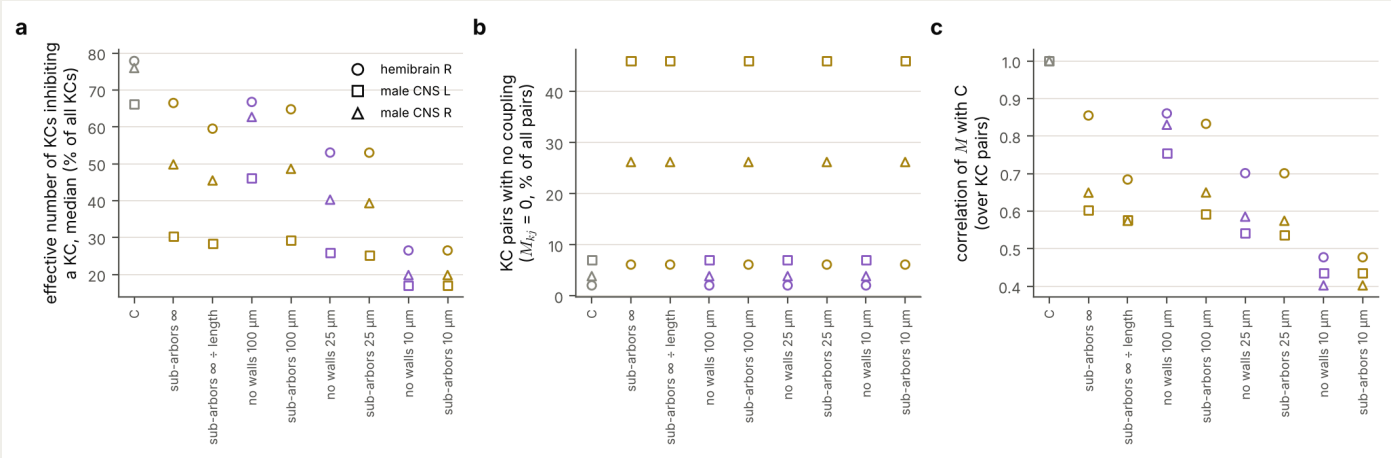


Figure 3. Walls cut couplings between compartments; at short λ they change nothing. Properties of the coupling matrix M (official ROIs), one marker per mushroom body. x-axis: versions; gray = C (one APL), purple = local without walls (Report 11), gold = confined to sub-arbors, at the same λ side by side. **a**, for each KC k the participation ratio $(\sum_j M_{kj})^2 / \sum_j M_{kj}^2$, the number of KCs that would contribute equally to its inhibition, as a % of all KCs; median over KCs. **b**, % of all KC pairs (k, j) whose coupling is exactly zero: with walls, two KCs with no compartment in common cannot inhibit each other. **c**, Pearson correlation of M with C over KC pairs. From `reports/data/r12_M_properties.csv`.

With isopotential compartments 6% of KC pairs in hemibrain R, 46% of KC pairs in male CNS L, 26% of KC pairs in male CNS R cannot interact at all, yet a typical KC is still inhibited by 66%, 30%, 50% of all KCs (same order), because its synapses lie in the big compartments. At λ = 10 µm the walled and unwalled matrices are practically the same (the signal has faded long before it would reach another compartment, 200–360 µm away), so from about 25 µm down this model and Report 11's coincide.

4 Results: rank order and pattern

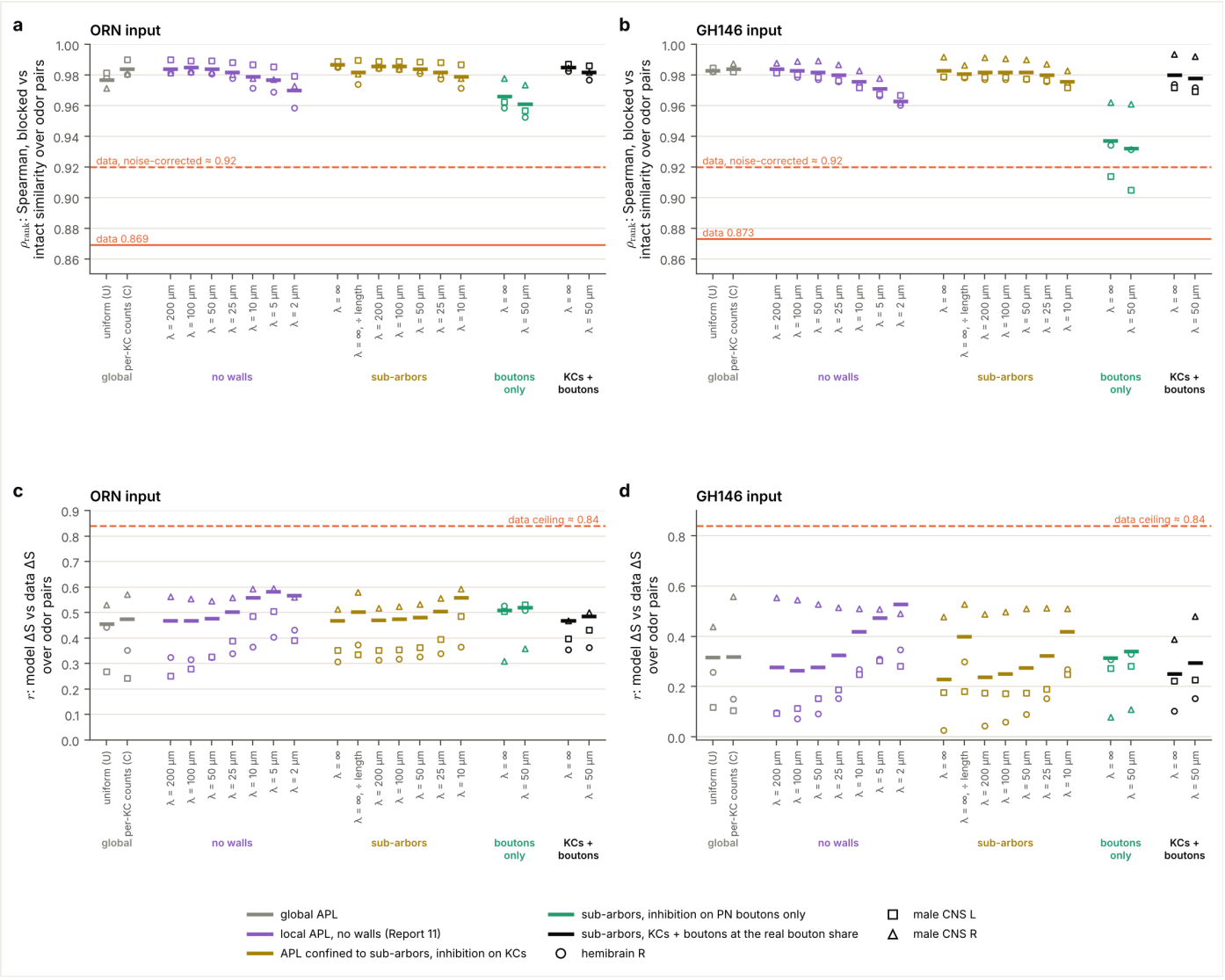


Figure 4. Confining APL to its sub-arbors does not reorder the pairs more. Official ROIs. x-axis: model versions in five groups (colors, named in the legend and under the axis): one global APL (U, C); the local APL without walls of Report 11 (λ from 200 to 2 µm); APL confined to sub-arbors with inhibition on KCs (λ = ∞, λ = ∞ with each compartment's potential divided by its cable length, then λ = 200 to 10 µm inside compartments); confined to sub-arbors with inhibition on PN boutons only; confined, with KCs and boutons at the real bouton share. Open markers: the three mushroom bodies; bar: their mean (in **c**–**d** the bar is r of the similarity change averaged over mushroom bodies, so it can lie above all three markers). **a**–**b**, rank order ρ_{rank}, lower = more reordering; orange lines: data, raw (solid) and noise-corrected (dashed). **c**–**d**, pattern r; orange dashed: the data's ceiling. Left: ORN input (16 odors, 120 pairs); right: GH146 input (13 odors, 78 pairs).

Table 3. Main numbers, official ROIs. Selected versions; all are in `APL_compartment_model_summary.json` and, for the versions without walls, `APL_local_model_summary.json`.

Input	Version	ρ_{rank} (mean of MBs)	pattern r	mean ΔS
ORN	uniform (U)	0.977	0.455	0.079
ORN	per-KC counts (C)	0.984	0.475	0.049
ORN	sub-arbors, isopotential (λ = ∞)	0.987	0.469	0.042
ORN	sub-arbors, isopotential ÷ cable length	0.982	0.503	0.047
ORN	sub-arbors, λ = 50 µm inside	0.984	0.481	0.043

Input	Version	ρ_{rank} (mean of MBs)	pattern r	mean ΔS
ORN	no walls, λ = 50 µm (Report 11)	0.984	0.477	0.043
ORN	sub-arbors, λ = 10 µm inside	0.979	0.558	0.044
ORN	no walls, λ = 10 µm (Report 11)	0.979	0.558	0.044
ORN	no walls, λ = 2 µm (Report 11)	0.970	0.567	0.045
ORN	sub-arbors, boutons only (λ = ∞)	0.966	0.509	0.067
ORN	sub-arbors, KCs + boutons (λ = ∞)	0.985	0.469	0.046
ORN	**data** (TNT vs TNT3c)	**0.869** (≈ 0.92 noise-corrected)	ceiling ≈ 0.84	**0.178**
GH146	uniform (U)	0.983	0.316	0.061
GH146	per-KC counts (C)	0.984	0.319	0.037
GH146	sub-arbors, isopotential (λ = ∞)	0.983	0.230	0.034
GH146	sub-arbors, isopotential ÷ cable length	0.981	0.399	0.032
GH146	sub-arbors, λ = 50 µm inside	0.982	0.274	0.032
GH146	no walls, λ = 50 µm (Report 11)	0.982	0.278	0.032
GH146	sub-arbors, λ = 10 µm inside	0.976	0.418	0.029
GH146	no walls, λ = 10 µm (Report 11)	0.976	0.418	0.029
GH146	no walls, λ = 2 µm (Report 11)	0.963	0.528	0.029
GH146	sub-arbors, boutons only (λ = ∞)	0.937	0.315	0.053
GH146	sub-arbors, KCs + boutons (λ = ∞)	0.980	0.250	0.037
GH146	**data** (TNT vs TNT3c)	**0.873** (≈ 0.92 noise-corrected)	ceiling ≈ 0.84	**0.201**

Answer, official ROIs

Rank order: no. Confined to sub-arbors, inhibition on KCs leaves ρ_{rank} at 0.979–0.987 (ORN) and 0.976–0.983 (GH146), next to U (0.977, 0.983) and C (0.984, 0.984), and far above the data's ≈ 0.92. Inhibition on boutons only, confined to sub-arbors, reaches 0.966 (ORN), 0.937 (GH146) at λ = ∞, against 0.957, 0.931 without walls (Report 11): the walls do not make it reorder more. With KCs and boutons at the real share it is back to 0.985, 0.980.

Pattern: better than one APL only when small compartments count more, and never better than without walls. Isopotential compartments give r = 0.469 (ORN), 0.230 (GH146), no better than C; dividing each compartment's potential by its cable length gives 0.503, 0.399, against U 0.455 / 0.316, C 0.475 / 0.319 and the best local version without walls 0.581 (λ = 5 µm, ORN), 0.528 (λ = 2 µm, GH146). With decay inside compartments r rises as λ shrinks, but only to the values the same λ gives without walls.

Placement within compartments: shuffling KCs within class for ORN (λ = ∞, ÷ length), GH146 (λ = ∞, ÷ length) gives ORN: r 0.503 vs shuffles 0.420, p = 0.13; GH146: r 0.399 vs shuffles 0.233, p = 0.01.

Size: no. Mean ΔS of the KC-route compartment versions is 24%–26% of the data's (ORN), 14%–17% of the data's (GH146), smaller than U's.

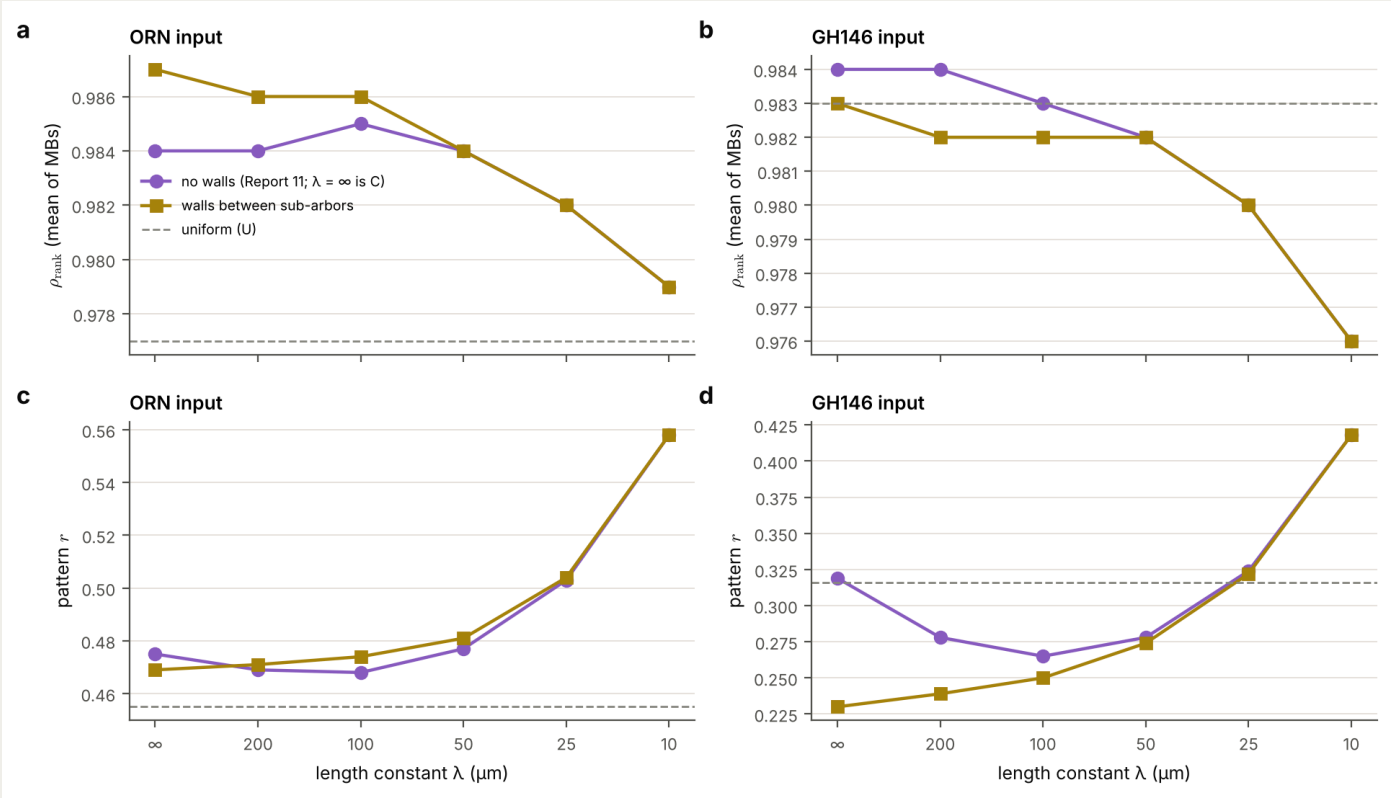


Figure 5. Walls versus no walls at the same length constant. The same length constant with and without walls (official ROIs). x-axis: λ, from ∞ (left; without walls this is C, with walls the isopotential compartments) to 10 μm. Purple circles: Report 11's local model; gold squares: the same with no signal crossing between sub-arbors; gray dashed: uniform APL. **a**–**b**, rank order ρ_{rank} (mean of the three mushroom bodies); **c**–**d**, pattern r (MB-averaged change). Left: ORN input; right: GH146 input.

The walls matter only where the signal would otherwise reach another sub-arbor, at long λ (∞ to 50 μm). There they change ρ_{rank} by −0.002 to +0.003 and r by −0.089 to +0.006 (mostly lowering it, most for GH146). At λ = 10 μm walled and unwalled r agree to three decimals.

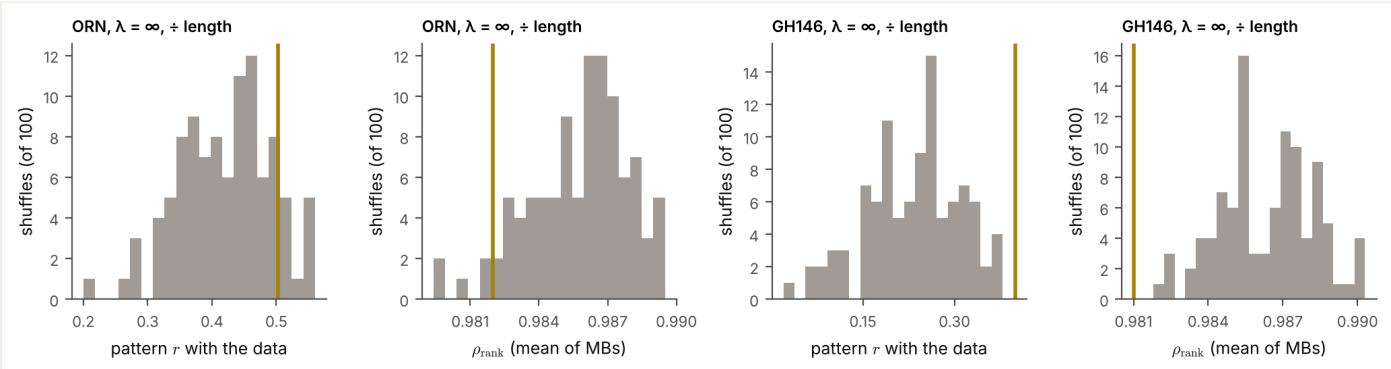


Figure 6. Does it matter which KC sits in which compartment? Gray: 100 shuffles in which KCs swap their APL synapse sites (and so their compartments) with other KCs of the same class, in all three mushroom bodies at once; gold line: the real wiring. Shown for the isopotential version with the better pattern r (panel titles). Left pair: ORN input; right pair: GH146 input; in each pair the pattern r (higher = closer to the data), then the rank order (lower = more reordering).

5 Second version: synapses outside the official ROIs added

The same model on the tables of `with_outside_ROI_synapses/`, which mostly adds the back of the male CNS calyx (Report 02). The compartments are the same skeleton pieces; more synapses fall into them.

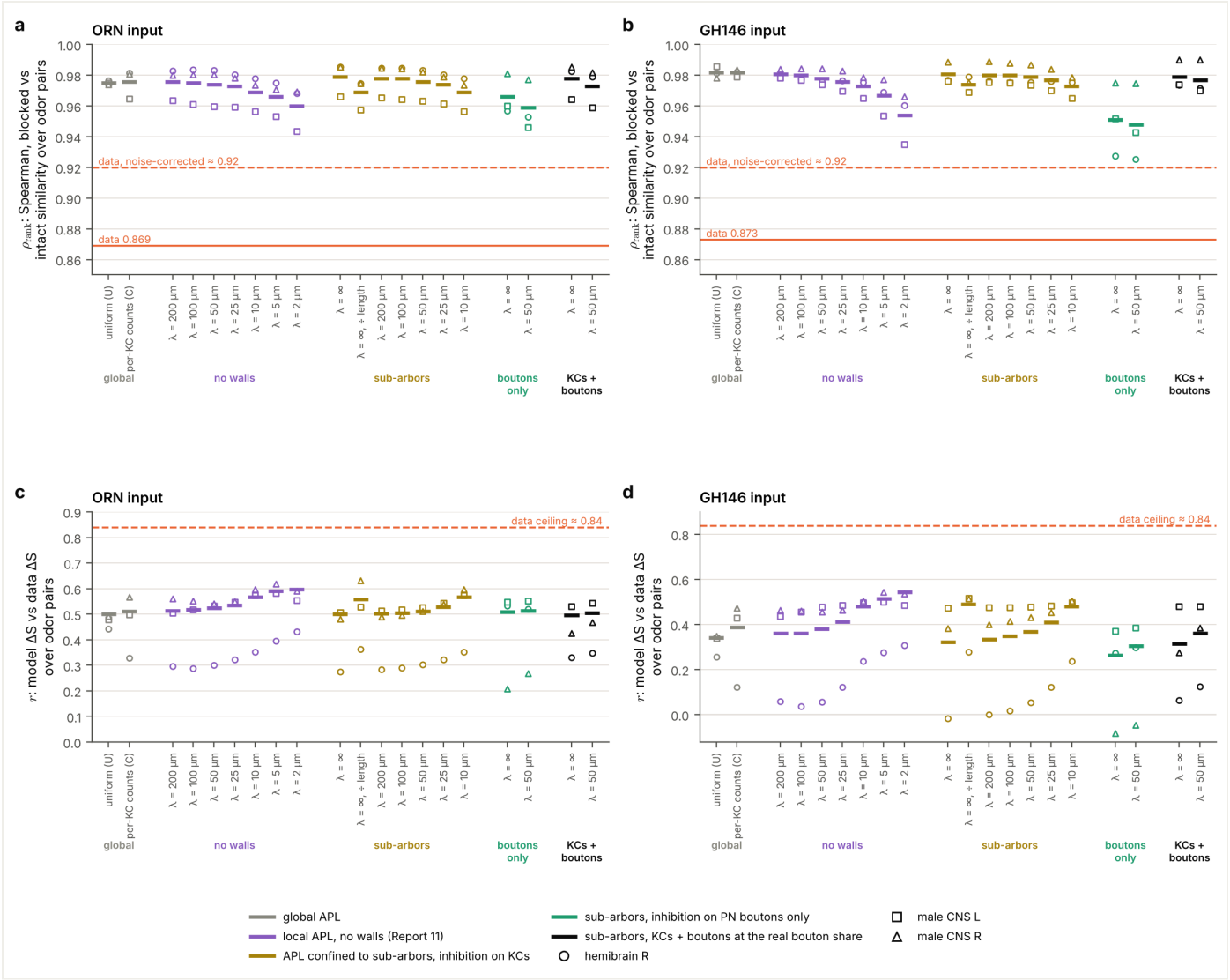


Figure 7. Rank order and pattern, second version. As Figure 4, with the tables of `with_outside_ROI_synapses/`.

Table 4. Official ROIs vs the second version. As Table 3.

Input	Version	ρ_{rank} official	ρ_{rank} second version	r official	r second version
ORN	uniform (U)	0.977	0.975	0.455	0.500
ORN	per-KC counts (C)	0.984	0.976	0.475	0.512
ORN	sub-arbors, isopotential (λ = ∞)	0.987	0.979	0.469	0.500
ORN	sub-arbors, isopotential ÷ cable length	0.982	0.969	0.503	0.558
ORN	sub-arbors, λ = 50 µm inside	0.984	0.976	0.481	0.511
ORN	no walls, λ = 50 µm (Report 11)	0.984	0.974	0.477	0.524
ORN	sub-arbors, λ = 10 µm inside	0.979	0.969	0.558	0.567
ORN	no walls, λ = 10 µm (Report 11)	0.979	0.969	0.558	0.568
ORN	no walls, λ = 2 µm (Report 11)	0.970	0.960	0.567	0.596
ORN	sub-arbors, boutons only (λ = ∞)	0.966	0.966	0.509	0.510
ORN	sub-arbors, KCs + boutons (λ = ∞)	0.985	0.978	0.469	0.496
GH146	uniform (U)	0.983	0.982	0.316	0.342
GH146	per-KC counts (C)	0.984	0.982	0.319	0.388

Input	Version	ρ_{rank} official	ρ_{rank} second version	r official	r second version
GH146	sub-arbors, isopotential (λ = ∞)	0.983	0.981	0.230	0.322
GH146	sub-arbors, isopotential ÷ cable length	0.981	0.974	0.399	0.490
GH146	sub-arbors, λ = 50 µm inside	0.982	0.979	0.274	0.369
GH146	no walls, λ = 50 µm (Report 11)	0.982	0.978	0.278	0.380
GH146	sub-arbors, λ = 10 µm inside	0.976	0.973	0.418	0.481
GH146	no walls, λ = 10 µm (Report 11)	0.976	0.973	0.418	0.481
GH146	no walls, λ = 2 µm (Report 11)	0.963	0.954	0.528	0.545
GH146	sub-arbors, boutons only (λ = ∞)	0.937	0.951	0.315	0.264
GH146	sub-arbors, KCs + boutons (λ = ∞)	0.980	0.979	0.250	0.314

Answer, second version

Rank order: no. Confined to sub-arbors, inhibition on KCs leaves ρ_{rank} at 0.969–0.979 (ORN) and 0.973–0.981 (GH146), next to U (0.975, 0.982) and C (0.976, 0.982), and far above the data's ≈ 0.92. Inhibition on boutons only, confined to sub-arbors, reaches 0.966 (ORN), 0.951 (GH146) at λ = ∞, against 0.961, 0.950 without walls (Report 11): the walls do not make it reorder more. With KCs and boutons at the real share it is back to 0.978, 0.979.

Pattern: better than one APL only when small compartments count more, and never better than without walls. Isopotential compartments give r = 0.500 (ORN), 0.322 (GH146), no better than C; dividing each compartment's potential by its cable length gives 0.558, 0.490, against U 0.500 / 0.342, C 0.512 / 0.388 and the best local version without walls 0.596 (λ = 2 µm, ORN), 0.545 (λ = 2 µm, GH146). With decay inside compartments r rises as λ shrinks, but only to the values the same λ gives without walls.

Placement within compartments: shuffling KCs within class for ORN (λ = ∞, ÷ length), GH146 (λ = ∞, ÷ length) gives ORN: r 0.558 vs shuffles 0.490, p = 0.09; GH146: r 0.490 vs shuffles 0.316, p = 0.01.

Size: no. Mean ΔS of the KC-route compartment versions is 31%–37% of the data's (ORN), 15%–18% of the data's (GH146), smaller than U's.

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6 What this means, and caveats

- Confining APL's signal to its calyx sub-arbors does **not** make inhibition reorder odor pairs more. The rank order stays at about 0.98, like a uniform APL. In this model, inhibition that acts on the KCs, whether global, local or split into compartments, mostly rescales KC responses and keeps the order of similarities (Report 09 explains when that happens).
- For **which** pairs change most, walls alone do not help: isopotential compartments match the data no better than one APL. Only when each compartment's potential is divided by its cable length, so that small compartments respond more strongly, does r rise above U and C, and for GH146 input that version beats within-class shuffles in both versions of the tables (for ORN it does not). It still stays below Report 11's local model without walls at short λ, and at short λ the walls are irrelevant because the signal fades before reaching another sub-arbor.

- The sub-arbors are uneven: hemibrain R is nearly one compartment, so most of any effect would come from male CNS. Most KCs have synapses in more than one compartment, which spreads each KC's inhibition over several compartments and blurs the walls.

Caveats

The sub-arbors come from the skeletons' connectivity. A connection missed by the segmentation, or a false merge, would change them (Report 10); worth checking in neuroglancer before relying on them.

Every small piece is treated as its own compartment. They hold 1–4% of APL's synapses, so pooling them differently would change little.

"÷ length" assumes the same branch thickness everywhere; real input resistance also depends on thickness and on the connections through the peduncle, which the walls ignore.

Synapse counts stand in for strength; a steady state is assumed; the shuffle p values do not correct for choosing between the two isopotential versions.

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